Financial Tweet Sentiment Prediction — Model Comparison Report

Deep Learning & BERT Project | RNN vs. LSTM vs. GRU baselines

1. Objective & Dataset

This project builds a 3-class sentiment classifier (Bearish / Bullish / Neutral) for finance-related tweets, using the Twitter Financial News Sentiment dataset (9,543 training tweets, 2,388 validation tweets after removing duplicates and 6 empty rows). The dataset is imbalanced (~65% Neutral, ~20% Bullish, ~15% Bearish), which was addressed using class-weighted cross-entropy loss (weights: Bearish 2.2, Bullish 1.65, Neutral 0.5) rather than oversampling/undersampling/SMOTE, since resampling techniques risk distorting raw text data.

2. Preprocessing Pipeline

Tweets were cleaned (URLs, mentions, quotes, punctuation and extra spacing removed; stock tickers such as $AAPL and negative values such as -5% preserved), lowercased, and tokenized by whitespace. A vocabulary of 19,501 unique tokens (plus <PAD> and <UNK>, total 19,503) was built **exclusively from the training set** to avoid data leakage; unseen validation words map to <UNK> (measured at 11.3% of validation tokens). Sequences were encoded to integers and padded/truncated to a fixed length of 20 tokens, chosen as the 95th percentile of training tweet lengths to balance information loss against wasted padding.

3. Model Architectures & Training Methodology

Three baseline architectures were implemented and compared, each using an Embedding layer (vocab size 19,503, embedding dimension 50) feeding into a single recurrent layer (hidden size 64) and a final Linear layer producing 3 class scores: **(1) Embedding + Simple RNN, (2) Embedding + LSTM, (3) Embedding + GRU**. All models were trained with the Adam optimizer (lr=0.001), batch size 64, using class-weighted cross-entropy loss. **Early stopping** (patience = 3 epochs, max 50 epochs) was used for every model: validation accuracy was checked after each epoch, the best-performing weights were retained, and training stopped automatically once validation accuracy failed to improve for 3 consecutive epochs. This replaced an earlier manual approach of picking a fixed epoch count, which produced inconsistent, unreliable results for Simple RNN in particular.

4. Results

Under early stopping, Simple RNN converged to a degenerate solution that heavily favors the majority Neutral class (Bearish/Bullish F1 near zero), consistent with its known difficulty retaining context over longer sequences (vanishing gradients). Both LSTM and GRU learned substantially more balanced, useful representations, with GRU outperforming LSTM on every metric in this run.

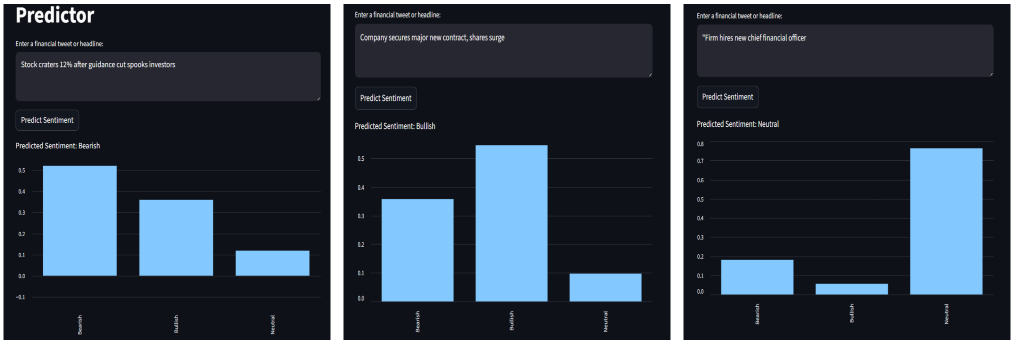
Model	Accuracy	Macro F1	Bearish F1	Bullish F1	Neutral F1
Simple RNN	63.5%	0.283	0.05	0.02	0.78
LSTM	68.8%	0.617	0.48	0.59	0.79
GRU (deployed)	74.5%	0.659	0.52	0.62	0.84

All figures from a single consistent early-stopping run across all three architectures (same patience, same max epochs, same data split).

GRU was selected as the deployed model, achieving the best accuracy and macro F1, and the most balanced performance across all three classes. Bearish remained the weakest class across every architecture, consistent with it having the fewest training examples (1,442 vs. 6,178 for Neutral) — suggesting the remaining gap is driven more by data scarcity than architecture choice. Simple RNN's collapse under a fair, automated stopping rule (rather than a manually-picked epoch count) is itself an informative result: an earlier manual run had coincidentally produced a much higher RNN score at one specific epoch count, which would have overstated RNN's true, generalizable capability had it been reported instead.

5. Error Analysis & Streamlit Dashboard

A Streamlit dashboard (below) accepts a free-text tweet/headline, cleans and encodes it using the same pipeline as training, and displays the predicted class plus a bar chart of class probabilities from the deployed GRU model. Manual testing surfaced two notable failure modes worth flagging: (1) the word "crash" was found to appear almost as often in Neutral-labeled training tweets as Bearish ones (10 vs. 9 occurrences), so a confident Neutral prediction for standalone phrases like "stock crash" reflects genuine ambiguity in the training labels rather than a modeling error; (2) for tweets longer than 20 tokens, truncation can remove decisive sentiment cues — a 24-token Apple earnings headline flipped prediction once the trailing phrase "shares up 4% in after-hours trading" was shifted inside vs. outside the retained 20-token window. With the final GRU model, the dashboard correctly classified varied Bearish, Bullish, and Neutral test headlines with clear, well-separated confidence scores, shown below.



Left to right: correct Bearish, Bullish, and Neutral predictions from the deployed GRU model, each with a clearly dominant probability bar for the correct class.

6. Conclusion & Future Work

GRU is the recommended baseline for this task under a fair, automated training regime, meaningfully outperforming both Simple RNN and LSTM. Future improvements could include: increasing max sequence length or truncating from a different position to reduce information loss on longer texts, collecting more Bearish-labeled examples to close the remaining class gap, and fine-tuning a pre-trained FinBERT model (attempted but descoped in this iteration due to CPU training time constraints) to test whether pre-trained language understanding improves on the GRU baseline.